

Design and Implementation of a SCADA Based Boiler Monitoring and Controlling System

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Abstract—The SCADA system offers a comprehensive perspective, and PLC technology is integrated for real-time monitoring and control. A focus on safety includes sensors that can react quickly to deviations and turn on devices like the steam release solenoid valve and feedwater supply pump. Potential damage is avoided by the proactive temperature control provided by a pump. Real-time visibility and prompt decision-making are ensured by SCADA. Large-scale data storage and SCADA operations are carried out via the Haiwell Industrial IoT Cloud platform. Terminals on a variety of platforms, such as PC, iPad, Android, iOS, and others, are supported by Haiwell Cloud. Haiwell Cloud ensures proactive monitoring, instantly notifying users via email if boiler temperatures surpass set limits. On-site buzzers provide real-time alerts for swift response. The system exhibits versatility since it is tailored for a wide range of industrial applications.

Keywords—SCADA Monitoring System, Sensor Based Monitoring System, Haiwell Cloud platform, Real time Alert.

I. INTRODUCTION

Approximately 3000 boilers, or roughly 60 Percent of all boilers in the nation, are used in Bangladesh's RMG industry. About two-thirds of these industries are situated in Gazipur and Narayanganj, two nearby districts of Dhaka. The nation is rapidly becoming more industrialized, which has led to an increase in the number of new boilers getting operating permits and a 27 Percent increase in the overall number of licensed boilers in the entire country over the last four years [1]. Over the past decade, 237 events have occurred in 4,560 businesses within Bangladesh's apparel industry. Building collapses accounted for 1.27 Percent, boiler breaks for 3.38 Percent, fires for 94.09 Percent, and miscellaneous events for 1.27 Percent. 6,870 lives lost 1,689 wounded and 5,181 killed were the result. There are urgent safety concerns after a dramatic surge from 2011 to 2015 that left 1,287 people dead and 3,800 injured [2]. This article uses a water level sensor to measure the levels in the tanks, and temperature sensors for each boiler to assure accurate management. A PIC

microcontroller, GSM modem, sensors, and interface circuits carry out the control operations, assuring effective and tailored automation for the intended results [3]. This study introduces an innovative automated boiler control system using a PLC to minimize errors. Temperature thresholds are used by the system to automatically start and stop the boiler. A sensor monitors temperature by sending signals to the PLC. Furthermore, the ON-OFF push button enables operation for a specified period of time [4]. This study presents a smart boiler system that uses LM35 temperature sensors, a buzzer to sound warnings, and Arduino as the main controller. Two LM35 sensors are used to measure temperature, and the Arduino is used to process the data. If the temperature rises above a threshold specified by a potentiometer, the buzzer will sound. Precise temperature monitoring is ensured by the functional temperature range of the LM35 sensors [5]. In this project, data is sent to the microcontroller via a temperature sensor for the boiler and a combined temperature and pressure sensor. Effective boiler control is ensured by the LCD's readings and buzzer alerts for variations in pressure and temperature in real time [6]. This system enhances fault detection in steam power plants by integrating a novel approach into PLCs, surpassing traditional hardware-based coding. It focuses on three major boiler failures, employing Allen Bradley PLC, and utilizes real-time SCADA visualization for comprehensive plant monitoring and control [7]. The Boiler Control and Monitoring System utilizes the NodeMCU ESP8266 Wi-Fi module for smooth communication. The ESP8266 MCU is used along with water level, ultrasonic, temperature, and humidity sensors to automate boiler sector monitoring in the prototype [8].

This innovative boiler automation project seamlessly integrates PLC technology and SCADA for real-time monitoring and control. With an emphasis on safety, it uses sensors to quickly react to deviations and turn on devices like the steam release solenoid valve and feedwater supply pump. While SCADA provides fast decision-making and real-time visibility,

proactive temperature control averts damage. It is made for a variety of industrial uses and provides a firm basis to pursue progress. Ensuring temperature control by PLC-triggered feed-water delivery and real-time SCADA monitoring, the system is activated at 30°C. An instantaneous PLC-managed shutdown takes place at 45°C, ensuring a regulated halt for increased dependability and safety. By averting extremes, this temperature-based approach enhances boiler automation's general safety and dependability. The system employs SCADA with PLC GXWorks2, HMI, and Haiwell industrial IoT cloud platform. The PLC and SCADA communicate through Modbus TCP/IP. Validation of system accuracy involves comparing HMI data with SCADA and Haiwell Cloud. The Haiwell Cloud app supports various platforms, ensuring easy remote monitoring and control. Its secure communication features include 128-bit SSL encryption, A-key/B-key, and password checks for device terminals, ensuring safe and stable remote access.

II. OVERVIEW OF THE SYSTEM

The block representation of the project is shown in Fig. 1. This project employs a Mitsubishi Fx3u 24MRES PLC, utilizing GXWorks2 Software to create a ladder diagram. Three sensors (temperature, pressure, level) feed data to the PLC, controlling components like the feed pump, heater on/off, steam outlet, and alarm. The SCADA system, linked to the PLC, is emulated using Haiwell cloud platform. An HMI interface allows data review, while the Haiwell cloud HMI Emulator facilitates simulation without physical devices. Emergency switches and a start/stop system enhance safety. Buzzers alert when temperatures exceed the predefined thresholds. Use wireless SCADA to transform control dynamics and provide smooth real-time monitoring and control. Our interface combines historical patterns, current data, and system status to present a holistic view. Have a strong alert system and respond quickly to anomalies.

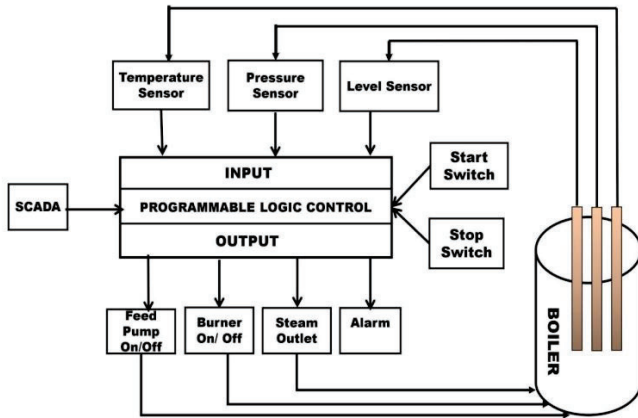


Fig. 1. Block Diagram of the Project.

III. METHODOLOGY

This section outlines the steps to create a prototype of the system. Data Transmission, Data Storage and SCADA visual

monitoring are all part of the development process for boiler monitoring and control. To effectively control boilers, PLC, SCADA, and Haiwell Cloud are used.

A. Circuit Diagram of the System

The boiler system, depicted in Fig. 2, is efficiently controlled by a Mitsubishi FX3U-24MRES PLC with 14 in-puts and 10 outputs. This PLC manages critical functions such as START (X0), STOP (X1), water levels (LOW=X2, MEDIUM=X3, HIGH=X4), steam temperature, and pressure switches. Notably, only 10 inputs and 4 outputs are utilized in this project. Inputs include START (X0), STOP (X1), Water Level (LOW=X2, MEDIUM=X3, HIGH=X4), Steam Temperature 01 (A0 - Analog IN), Auxiliary Steam Temperature 02 (A1 - Analog IN), Pressure Switch 01 (HIGH=X5, LOW=X6), Auxiliary Pressure Switch 02 (HIGH System Trip=X7). Outputs comprise Feed Pump (Y0), Heater (Y1), Steam Outlet (Y2), and Alarm (Y3). The circuit's clarity and detailed representation of switch states and conditions, such as water levels, provide insightful information on the system's operational aspects.

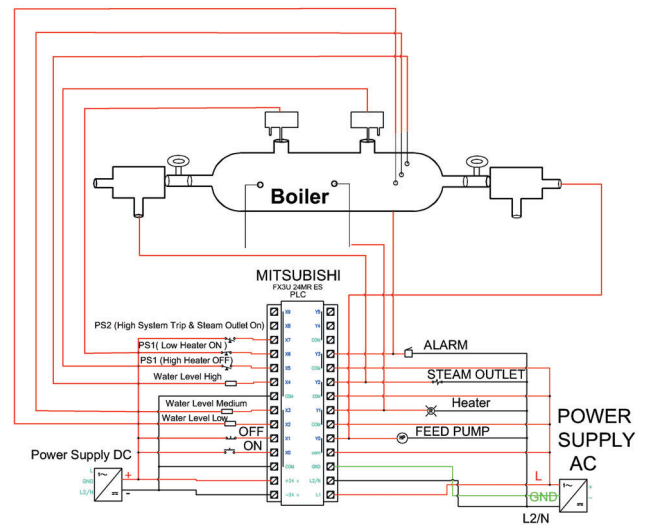


Fig. 2. Circuit Diagram of the Project.

B. Data Transmission

The Modbus TCP protocol is widely used in industrial control systems. Modbus-S Client and Modbus-S Server communication is a feature of Modbus TCP/IP. A net filter is used as a firewall for the Modbus TCP protocol gateway in order to increase security. It uses a White List assessment system to confirm the legitimacy of the data [9].

C. Website Data Storage and Alert

NET-Framework-based Haiwell Cloud SCADA is a platform for managing and monitoring industrial automation. Real-time and historical data can be transmitted across databases such as MySQL and SQL Server by using the

Haiwell data-link tool. Alarm displays and buzzer warnings are used to immediately transmit alarm information to the SCADA during operation.

D. PLC Ladder Diagram Design

In Fig. 3, memory addresses M800, D424, and D430 are used for temperature in Celsius. To set the temperature, the analog command RD3A is used. K0 represents channel 1, while K1 represents channel 2. The set value is a float number. For sensors, memory addresses M40, M48, and M52 are utilized. The START input is X000, and the STOP input is X001.

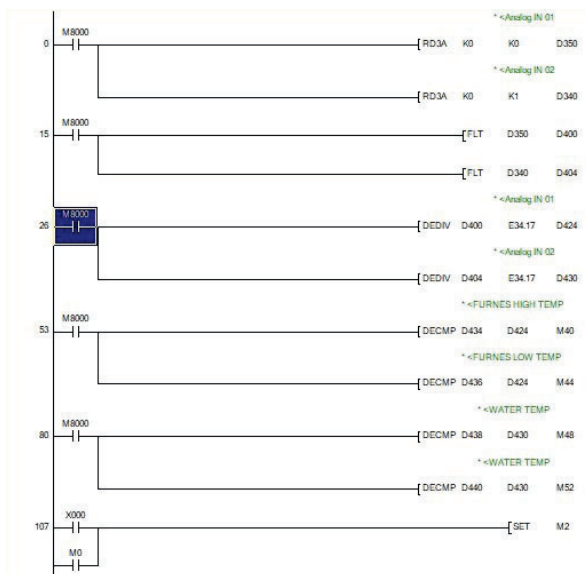


Fig. 3. Ladder part A of the Project.

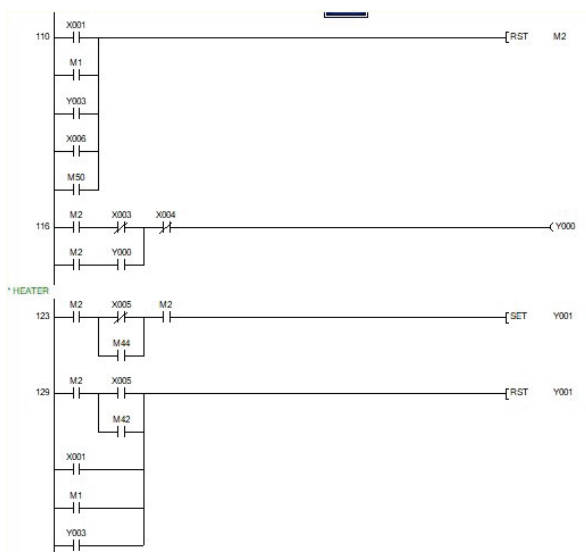


Fig. 4. Ladder part b of the Project.

In Fig. 4, set or reset commands for water level (medium X003, high X004), high and low setpoints from SCADA/Temp M2, pressure switches 01-X005, and 3202-X006 are depicted. Corresponding output commands for pump Y000, heater Y001, and steam outlet Y002 are also shown.

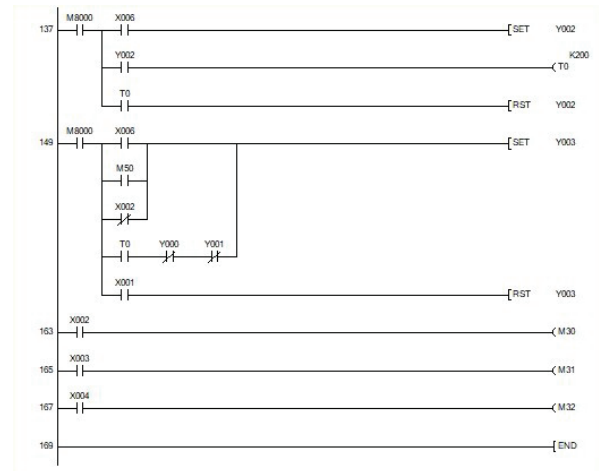


Fig. 5. Ladder part c of the Project.

In Fig. 5, set or reset command for low water level (X003) latched with pressure switch 02-X006 is depicted, with the corresponding output command to alarm Y003.

IV. PROGRAMMING, IMPLEMENTATION AND RESULTS

A. Flowchart

The boiler system control process is smoothly orchestrated by our project's operational approach, as illustrated in Fig. 6. The water level, temperature, and steam pressure are evaluated against calibrated values when the PLC and SCADA activate both systems. In accordance with high, medium, or low ratings, the burner, safety lamp, and pump motor operate. Alerts for surpassing temperature or pressure restrictions, as well as system tripping, are safety measures. Accurate mea-

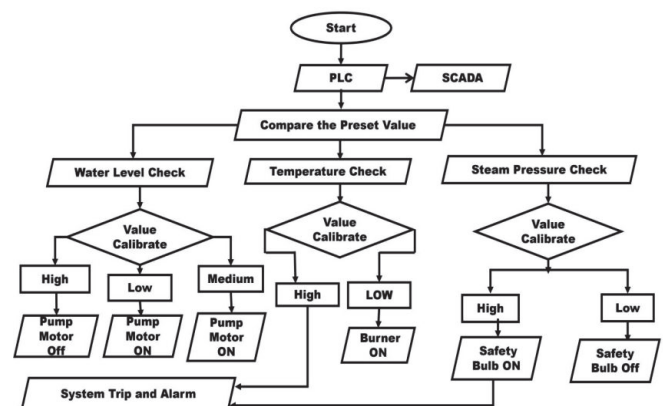


Fig. 6. Flowchart of the Project.

measurements are guaranteed through PLC and SCADA activation, concurrent parameter verification, and calibrated changes. Low water levels cause the pump motor to turn on, while high water levels cause it to turn off. After heating up to a certain temperature, the hob deactivates. Above a certain pressure threshold, the safety valve lets go of extra steam. Data processing, distribution, and visualisation are made possible by SCADA, that saves all system data.

B. Result & Analysis

The Overview of the project are depicted in Fig. 7. The Mitsubishi FX3U-24MRES PLC is included in this system together with SMPS, an indicator, a solenoid valve, a buzzer, temperature, water level, pressure, and magnetic conductor sensors. It uses the Modbus TCP/IP communication protocol to provide accurate temperature management. At 30°C, it initiates feedwater delivery, and at 45°C, it starts a PLC-managed shutdown for further safety. Real-time monitoring is made possible via the Haiwell IoT cloud and SCADA solution. By comparing data from HMI, SCADA, and Haiwell Cloud, system accuracy is verified, ensuring stability. Because of its strong security features, the Haiwell Cloud app allows for remote monitoring and control, which improves the boiler automation system's overall dependability and safety. Alarm displays and buzzers immediately transmit information to the SCADA system during operation.

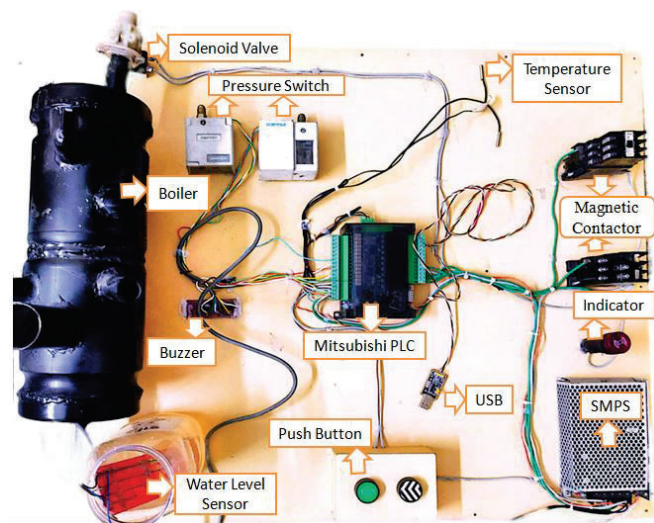


Fig. 7. Overview of the Project.

In Fig. 8, the simulation illustrates the initial state of the boiler system. The system activates when the temperature reaches or exceeds 30°C, and the voltage in the analog input A1 of the Mitsubishi Fx3u 24MRES PLC is at 1.5V. The boiling of water in the boiler initiates, leading to an increase in steam pressure.

In Fig. 9, a simulation illustrates the operating state of the boiler system. The system shuts off when the temperature reaches or exceeds 45°C, and the voltage in the Mitsubishi

Fx3u 24MRES PLC analog input A0 is at 1.5V. The water level drops, the burner stops, and the steam level is high.

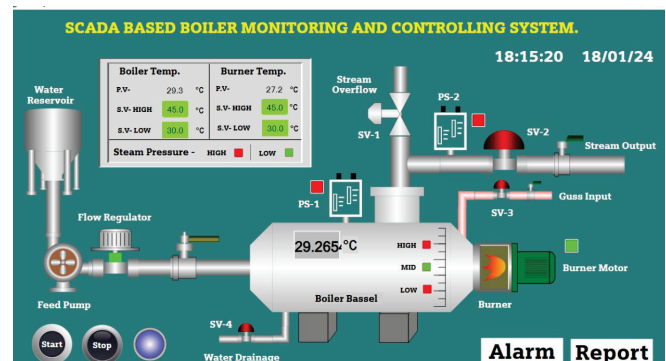


Fig. 8. The simulation of starting state.

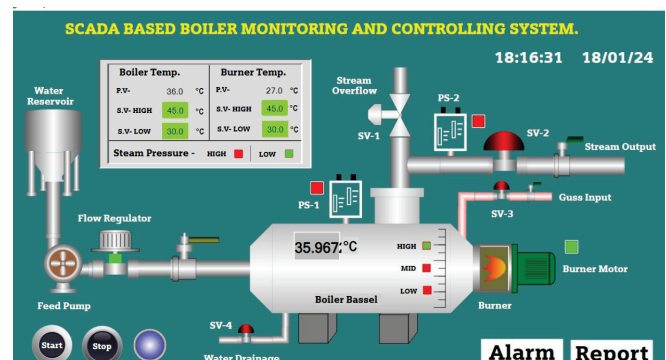


Fig. 9. The simulation of operating state.

In Fig. 10, the simulation depicts the stop state of the boiler system. The system automatically shuts down for additional safety when the temperature exceeds 45°C.

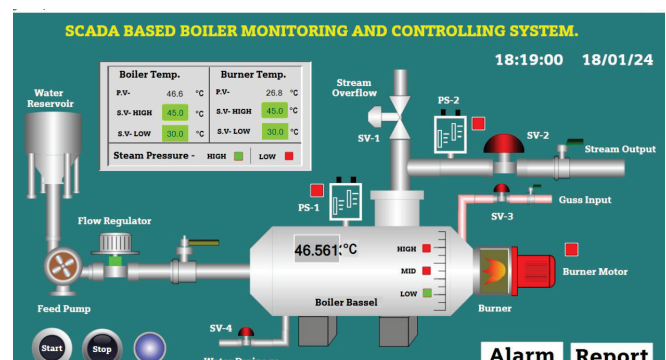


Fig. 10. The simulation of stop state of boiler system.

Reports on operations are generated in real time by Haiwell Cloud. A representation of the operating history data is presented in Fig. 11, which demonstrates the way the date, boiler temperature, and burner temperature have evolved over time. This visual aid is integrated to assist operators analyse performance.

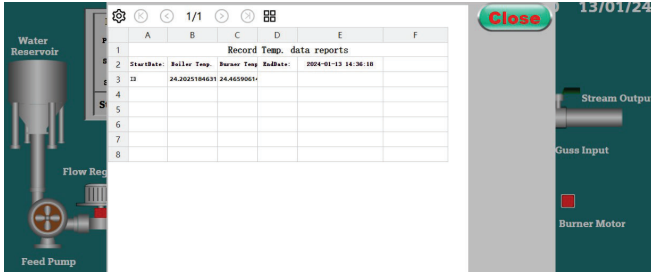


Fig. 11. Haiwell Cloud Real time data Visualisation.

The cloud-generated data table is depicted in Fig. 12. The times and dates specified in the table are altered in real time within specified intervals. Users are able to analyse real-time project data from SCADA with the assistance of this data.

Record Temp. data reports					
Boiler Temp.	Burner Temp.	Date	Time	Pump	Burner
28.91425323	26.95347023	2024-01-18	17:09:19	true	true
38.1914	28.7094	2024-01-18	17:09:20	true	true
38.77671	28.76793	2024-01-18	17:09:21	true	true
39.83026	28.82646	2024-01-18	17:09:22	true	true
40.59116	28.88499	2024-01-18	17:09:23	FALSE	true
40.85455	28.94352	2024-01-18	17:09:24	FALSE	true
41.14721	29.03131	2024-01-18	17:09:25	FALSE	true
41.46913	29.03131	2024-01-18	17:09:26	FALSE	true
41.73252	29.08985	2024-01-18	17:09:27	FALSE	true
42.20076	29.11911	2024-01-18	17:09:28	FALSE	true
42.81534	29.17764	2024-01-18	17:09:29	FALSE	true
43.57624	29.23617	2024-01-18	17:09:30	FALSE	true
44.83465	29.2947	2024-01-18	17:09:31	FALSE	true
45.12233	29.3825	2024-01-18	17:09:32	FALSE	FALSE

Fig. 12. Real-Time Data Table.

V. COST ANALYSIS

We described our research project's financial evaluation in this section. The hardware costs associated with the study are listed in Table I. By integrating components onto a

TABLE I
APPRAISAL OF THE SUGGESTED SYSTEM'S COST

No	Component Name	Quantity	Price (BDT)
1	Mitsubishi FX3U-24MRES PLC	1	12000/=
2	Temperature sensor	1	200/=
3	Pressure switch	2	1000/=
4	Solenoid valve	1	500/=
5	SMPS	1	400/=
6	Water level sensor	1	150/=
7	Magnetic contactor	2	1100/=
Total Installation Cost			15,350/=
Miscellaneous Cost			600/=
Total Cost			15,950 BDT ≈ \$145 (USD)

PCB board, a shift to commercial production could lower the estimated cost of the current prototype, which is currently estimated to be around 15950 BDT, to about 14000 BDT. Even though these figures suggest possible cost-effectiveness, a thorough feasibility study is necessary.

VI. CONCLUSION

The boiler system is an instance of an innovation in industrial control, utilising SCADA and emphasising real-time monitoring, current security technologies, and increased efficiency. Activating at 30°C and assuring a secure shutdown at 45°C, our system combines real-time SCADA monitoring and PLC-triggered feedwater supply to achieve exact temperature control. With Haiwell Cloud, users are able to simulate PLC-SCADA communication in Modbus TCP/IP and have proactive monitoring with real-time email warnings for temperature deviations. Buzzers placed on-site ensure quick response, as well as SCADA and large-scale data storage are made easier by the Haiwell Industrial IoT Cloud platform. By supporting several platforms, Haiwell Cloud improves overall system dependability and security by ensuring accessible on PC, iPad, Android, iOS, and etc. With its adaptability, durability, and responsiveness, it redefines the industrial environment and moves fluidly between manual and digital modes. This technology is a pioneer in industrial and power plant automation, providing unparalleled long-term cost-effectiveness, less downtime, and enhanced safety standards—all at a higher initial cost. With its advanced data visualisation, in-depth analysis, and massive history logging, it generates enormous cost savings, optimises operations, and proactively prevents equipment issues. This method is further enhanced by the inclusion of machine learning. It sets a new standard with remote accessibility and comprehensive analytics, offering cost-effective boiler control solutions for the industrial era.

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